MANIPAL UNIVERSITY JAIPUR

SCHOOL OF COMPUTER SCIENCE & ENGINEERING

SOFTWARE ENGINEERING LAB

LABORATORY RECORD – EXPERIMENT 1

IDENTIFYING REQUIREMENTS FROM THE PROBLEM STATEMENT

Research Collaboration and Management System

Course Code: CSE3132

Submitted by: Ishpreet Kaur

Section: I

Reg. No.: 2427030110

Date: 21/8/2026

Experiment – 1

- **Aim**

  To identify and analyse the requirements from the problem statement for a Research Collaboration and Management System.

- **Problem Statement**

  The Research Collaboration and Management System is a centralized platform for managing research projects, collaborations, documents, tasks, publications, opportunities, communication, notifications, and research activities. It is intended to provide a common platform for students, researchers, faculty members, supervisors, external collaborators, and administrators.

1. Hardware Interfaces

- Computer/Laptop
- Minimum 4 GB RAM
- Dual-core processor or higher
- Minimum 10 GB available storage
- Keyboard and mouse
- Internet connection
- Display resolution of at least 1366 × 768 recommended

2. Software Interfaces

- Windows 10/11, Linux, or macOS
- Web browser such as Google Chrome, Mozilla Firefox, or Microsoft Edge
- Visual Studio Code or any suitable IDE
- Database management system such as MySQL/PostgreSQL
- Git/GitHub for version control
- StarUML/draw.io for UML modelling

3. System Requirements

- **User Management:** Users can register, log in, and manage their research profiles and interests.
- **Project Management:** Researchers can create projects, add members, assign tasks, and track progress.
- **Researcher Collaboration:** Users can search for researchers, find collaborators, and send collaboration requests.
- **Research Opportunities:** Users can post and search for internships, projects, grants, conferences, and other opportunities.
- **Document Management:** Users can upload, share, and manage research papers, reports, datasets, and other documents.
- **Task Management:** Project members can create, assign, update, and complete research tasks.

- **Communication:** Users can communicate through messages, comments, and project discussions.
- **Publication Management:** Researchers can add and manage their research publications and related details.
- **Notifications:** Users receive notifications about collaboration requests, tasks, deadlines, and project updates.
- **Search and Filtering:** Users can search researchers, projects, and opportunities using relevant filters.
- **Analytics and Reporting:** The system can generate reports on projects, publications, collaborations, and research activities.
- **Security and Privacy:** The system protects user information, research data, and documents through controlled access.
- **Scalability:** The system should support an increasing number of users, projects, and research activities.
- **Feedback:** Users can provide feedback to help improve the system.

4. Identification of Major Users / Actors

- **Researcher / Student:** Creates profiles and projects, searches researchers and opportunities, joins projects, manages tasks, uploads documents, manages publications, tracks progress, and communicates with collaborators.
- **Faculty / Supervisor:** Creates research projects, manages teams, assigns tasks, monitors progress, reviews documents, provides feedback, manages publications, and generates reports.
- **External Collaborator:** Creates a profile, searches researchers and opportunities, sends collaboration requests, joins approved projects, shares documents, and communicates with research teams.
- **Administrator:** Manages users, research projects, system activities, system data, and reports.

5. Functional Requirements Identified

- FR1 – User Registration and Login: The system shall allow users to register and securely log in.
- FR2 – Profile Management: The system shall allow users to create, view, and update their research profiles.
- FR3 – Researcher Search: The system shall allow users to search for researchers according to research interests, skills, expertise, and institution.
- FR4 – Research Project Creation: The system shall allow authorized users to create research projects and specify project details.
- FR5 – Research Project Management: The system shall allow project owners to update project details, members, deadlines, and status.
- FR6 – Collaboration Requests: The system shall allow users to send, accept, or reject collaboration requests.

- FR7 – Research Team Management: The system shall allow project owners or supervisors to add and remove members from research teams.
- FR8 – Task Management: The system shall allow users to create, assign, update, and complete research tasks.
- FR9 – Document Management: The system shall allow authorized users to upload, view, share, and manage research documents.
- FR10 – Progress Tracking: The system shall allow users to monitor research project progress and milestones.
- FR11 – Publication Management: The system shall allow researchers to add and manage publication information.
- FR12 – Communication: The system shall allow research team members to communicate and exchange project-related information.
- FR13 – Notifications: The system shall notify users about relevant research and project activities.
- FR14 – Search Opportunities: The system shall allow users to search and filter research opportunities.
- FR15 – Reports and Analytics: The system shall generate reports related to research projects and activities.

6. Non-Functional Requirements Identified

- **Performance:** The system should provide an acceptable response time for normal user operations.
- **Security:** The system should provide secure authentication, authorization, password protection, role-based access, and secure document access.
- **Usability:** The system should have a simple and intuitive interface.
- **Reliability:** The system should reliably store and retrieve research data and documents.
- **Scalability:** The system should support increasing numbers of users, projects, documents, and collaborations.
- **Maintainability:** The system should have a modular architecture so that future features can be added easily.
- **Availability:** The system should be available to authorized users whenever research activities need to be performed.
- **Privacy:** The system should protect private research information and restrict unauthorized access.

7. Scope of the System

- User registration and authentication
- Researcher profile management
- Research project management
- Research team management
- Collaboration requests
- Research opportunity management

- Document management
- Task management
- Project progress tracking
- Communication
- Publication management
- Notifications
- Search and filtering
- Analytics and reporting

8. Expected Outcome

The identified requirements provide a clear understanding of the users, system functions, interfaces, security needs, and quality attributes of the Research Collaboration and Management System. These requirements form the basis for preparing the Software Requirement Specification (SRS) and UML models for the system.

○ Conclusion

The requirements of the Research Collaboration and Management System were identified and classified into hardware interfaces, software interfaces, functional requirements, non-functional requirements, system scope, and major user roles. The analysis establishes the foundation for subsequent software engineering activities such as SRS preparation and UML modelling.